

A Systematic Literature Review of Cloud computing Adoption and Impacts among Small Medium Enterprises (SMEs)

Noor Afzan Salleh
Dep. of Information Systems
International Islamic
University Malaysia
Kuala Lumpur, Malaysia
afzan.live@iium.edu.my

Husnayati Hussin
Dep. of Information Systems
International Islamic
University Malaysia
Kuala Lumpur, Malaysia
husnayati@iium.edu.my

Mohd. Adam Suhaimi
Dep. of Information Systems
International Islamic
University Malaysia
Kuala Lumpur, Malaysia
adam@iium.edu.my

Asma Md Ali
Dep. of Information System
International Islamic
University Malaysia
Kuala Lumpur, Malaysia
sis_asma@iium.edu.my

Abstract—Although cloud computing is one of the most significant trends in information technology acquisition today, its adoption amongst the SMEs is still behind the larger counterparts. Additionally, among those that use, many face challenges to gain benefits as what is normally claimed. More research is needed to understand the issue. The purpose of this paper is to present the findings of a Systematic Literature Review (SLR) conducted related to cloud computing adoption among SMEs, particularly focusing on the post adoption stage. SLR method was employed as this method enable the review been done in a more comprehensive and rigorous manner. A total of 39 relevant articles were reviewed and the findings indicate that most past researches on cloud computing and SMEs focused on adoption, exploring factors that affect the adoption. Very few studies looked at the post adoption stage or the impacts of cloud computing on SMEs.

Keywords—Cloud computing, Systematic Literature Review (SLR), Small and Medium Enterprises (SMEs), Post Adoption

I. INTRODUCTION

Cloud computing adoption is becoming more common nowadays as it represent an attractive IT acquisition option for many business organizations, both big and small. Nevertheless, small and medium-sized enterprises (SMEs) are generally lagging behind the larger counterparts in its adoption despite it offers cost effective solution to SMEs. SMEs are dissimilar from big companies in many ways, such as, they use more owner centric approach, have informal organizational structure, and more resource-constraint. These characteristics may account for the difference in cloud computing adoption level among SMEs compared to the larger enterprises.

According to the official National Institute of Standards and Technology (NIST) definition “*cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management*

effort or service provider interaction” [1]. Cloud computing service providers deliver all IT services on demand and the payment for the same has to be made for the real usage of the computer facilities and resources [2]. It can be the fitting of services in covering insufficient IT capability of SMEs at a relatively lower cost.

Many past researches have studied cloud computing use among businesses, however, most focused on larger organizations. Studies that look at computing use in the context of SMEs are emerging such as [3], [7], [8], [9]. Nevertheless there are still many unclear issues related to SMEs use of cloud computing that warrant more detailed study including the factors that influence adoption, challenges faced by SMEs in adoption cloud computing, the success level and the impacts of such use. Therefore, there is a need to further investigate cloud computing use by SMEs and the factors that influence their adoption or lack of adoption decision. Following this, a good start is to try to understand what others have studied related to this phenomenon. Therefore there is need to review past literature to get understanding about the extent of research work carried out in cloud computing and SMEs to identify the research gaps.

The purpose of this paper is to present a review of literature on cloud computing and SMEs, covering the adoption and post adoption stages. The Systematic Literature Review (SLR) technique is used as it is claimed that this technique enable relevant literature being reviewed in a rigorous and comprehensive manner. While there have been a few review papers produced on cloud computing and SMEs in the past, this SLR paper focus on both adoption and post-adoption stage while previous ones mostly focused on conceptualization of cloud computing and just adoption issue. In order to achieve the comprehensive protocol by [4] for conducting SLR, the details of the method will be discussed in the next section.

II. SLR METHOD

This study adopts the SLR technique proposed by Kitchenham's et al. methodological protocols [4]. Thus, conducting a systematic review is systematized via three main stages: planning, conducting and reporting. The milestone in systematic literature reviews is the explicit and the clear definition of a review protocol in the planning phase that guides its execution. It aims to reduce researchers' bias and helps in structuring the retrieved results.

A. Planning the Review

In this paper, [4]'s guidance of conducting SLR was adopted. At the beginning, the researchers identify the need of the review. Then, develop review protocol that contain seven protocol namely define research questions, define source of research, define criteria for inclusion an exclusion, define selection procedures and define data extraction procidures. This plan review protocol for conducted a SLR on Cloud computing studies in SMEs area has been develop prior to performing for this study. In this paper, the review protocol was conducted by one of the authors of this research and was validated by two senior researchers.

B. Defining the SLR's research questions

For defining the research question, we use the common method known as Population, Intervention, Comparison, Outcomes and Context (PICOC) [11] and supported by the systematic chronology for cloud computing research and SMEs. The summary of PICOC used for this study is described in Table 1.

Table 1: Summary of PICOC

Population	SMEs owner, SMEs Manager
Intervention	Adoption, Use
Comparison	N/A
Outcomes	SMEs performance
Context	Review(s) any empirical and conceptual study involved in CC adoption and performance among SMEs. No restriction on the type of study applied

Guided by the above, this SLR will answer comprehensively the following four research questions, as shown in Table 2.

Table 2 : SLR research questions

ID	Research Question	Main motivation
RQ1	What is the research purpose of the selected literature?	To identify the central issues explored or examined by past work on cloud computing and SMEs
RQ2	What theory or framework is used in the selected literature?	To identify the theoretical underpinnings of past literature on loud computing and SMEs
RQ3	What are value or benefits cloud computing for SMEs?	To identify the cloud computing benefits and values for SMEs
RQ4	What are cloud computing challenges?	To identify cloud computing challenges faced by SMEs

C. Identifying the Relevant Literature

To define and describe the accurate and appropriate search strings (terms used for searching), this study used a two-step approach proposed by [4] for optimal result as discussed below:

Step 1: Search was done based on keywords, synonyms and alternate words in the study questions. Guided by PICOC, the relevant search terms were identified. Considering the synonyms and abbreviations, the following keywords are classified as shown in Table 3.

Table 3:Terms derived based on synonym words

General Terms	Technological and business model terminologies
Cloud computing	Cloud services, virtualization, Clouds, SaaS, cloud-based enterprise, cloud ERP, Application Service Providers (ASPs).
Small Medium Enterprises (SMEs)	Small Medium Business (SMBs) ,Small firms, Small enterprises, small and medium firm, Small Medium and Micro Enterprise (SMME)
Use	adoption, diffusion, acceptance, migration
Performance	Post-adoption, extended usage, effectiveness, implementation, successful, return on investment (ROI), value, competitive advantages

Step 2: Search done on previous cloud computing and SMEs studies conducted using SLR. The search on titles, research keywords and general terms from step one were taken for the term derivation, as shown in Table 4.

Table 4: Terms derived from keywords found from related studies

Author	Year	Period Sampling	Paper Analyzed
Martens et al. (2011)	2011	2007-2010	40 Scientific
Yang and Tate (2012)	2012	2008-2011	205 publicatios
Hoberg et al. (2012)	2012	Not specified	60 publications
Walther et al. (2012)	2012	2004-2011	36 SaaS publications

Finally, with the search strings followed, this study uses a comprehensive search for past studies:

(Cloud computing OR cloud OR cloud ERP OR SaaS) AND (SMEs OR Firm Organization AND (adoption OR use) AND performance OR competitive advantages)

In order to get quality research, five databases were use in a wide-ranging search in the context of cloud computing phenomenon and SMEs (i.e. EBSCOhost, ACM Digital library, ScienceDirect, SpringerLink and IEEE). Then, the search string was repeated using the Google Scholar. Next step, all articles were filtered using Mendeley software based on inclusion and exclusion criteria. A total of 50 publications were imported into the Mendeley Library and considered as primary studies.

D. Selection of primary studies

The purpose of classifying the inclusion and exclusion criteria was to ensure that only relevant articles would be used in this study. All primary studies were carefully processed by using the criterion for inclusion and exclusion as shown in Table 5.

Table 5: Inclusion and exclusion criteria

ID	Specific criteria for evaluating the report	Score
S1	Is the research clearly focus on cloud computing and SMEs adoption?	Yes/No/ Partially
S2	Is the evaluation of cloud computing and SMEs clearly described?	Yes/No/ Partially
S3	Is the evaluation approach clearly described?	Yes/No/ Partially
S4	Is the evaluation result analysed or discussed?	Yes/No/ Partially
S5	Are limitations and future implications for cloud computing and SMEs clearly positioned?	Yes/No/ Partially

E. Quality Assessment

The purpose of quality assessment is to guide the interpretation of the findings and determine the strength of the inferences of the selected studies and considered as crucial. To evaluate the quality of the study requires validity based on the validity of the scope of the study, research publication and comprehensiveness. Therefore, this study is evaluated through its report. According to [5], to assess the quality of reporting in the publication of past studies, there are ten criterias of the reporting structure that is divided into two categories namely general criteria and specific criteria. General criteria are for reporting quality and specific criteria for evaluation of the reporting. The general criteria are generic in nature which include clarity in research problem, research objectives, and research methodology, sufficient literature reviewed and results are reported clearly. The specific quality assessment questions used in this review study is shown in Table 6.

Table 6: Specific criteria for evaluating the report

Include Criteria	Exclude Criteria
Organizational Level Study Qualitative, Quantitative, Mixed-Method Research published started year 2012- year 2017	Individual Level study Does not include research published before the year 2011
Language English only Full text paper	Other languages not accepted Shorter than 6 pages, editorials, abstracts and No- peer reviewed studies
SME context only	Field or area other than SMEs or organization

Based on this ten criteria, 39 papers got scored 1, while 11 papers only scored 0.5 because of lack of guidance of review methodology and result analysis. For the purpose of review analysis, only 39 papers were used.

III. REVIEW RESULTS

This section presented the evidence of the SLR. Here, SLR procedures involved three main activities namely: i) quick scanning based on the automated search; b) scrutiny; and c) manual reference snowballing. As an evidence, the entire procedure of identifying and selecting primary studies are discussed below:

a) *Quick scanning* $n = 260$.

b) *Scrutiny* $n = 44$

The idea of the manual search was to gain confidence in the comprehensiveness of the search results. The scrutiny process excluded irrelevant studies or duplicate papers.

c) *Snowballed publication* $n = 6$ papers

This process of pursuing references of references is known as snowballing as suggested by [6]. Subsequently, these 6 articles [7]–[12] were retrieved via Google Scholar and added to Mendeley to prepare the pre-final set of primary studies. Finally, 50 articles were identified as the final set of primary studies and formed the basis for synthesis of evidence.

IV. SYNTHESIS OF EVIDENCE

In this section, details of the results of the SLR research questions and the synthesis of cloud computing and SMEs literature are discussed. It requires the essential to review and the ability to structure, write, and present the findings in a clear, solid and reliable manner. Hence, it will show the synthesized acquaintance knowledge of the accumulated facts in the research area. In the context of this study, to obtain a worthy review it is guided by four identified research questions,

referred to as RQ1, RQ2, RQ3 and RQ4. Answers to these research questions are achieved by reviewing 50 published articles about cloud computing and SMEs by sampling a five year period (2013-2017). This following section will discuss in detail the essence of the SLR findings realted to RQ1-RQ4.

A. RQ1: What is the research purpose of the selected literature?

There has been a notable increase of cloud computing research over the years, and it continues to develop to satisfy the businesses' requirement. The content of articles that are published in scholarly journals can give an idea about the change in research and themes about cloud computing. The purpose of RQ1 is to set light on the progress of cloud computing research trend in the SMEs context. When reading the single studies for purposes of excerpting or extract, we engages in three salient steps called open coding, axial coding, and selective coding to develop classification scheme. Thus, by classification of the articles, it takes a holistic approach to review and better understand the purpose of the literature on cloud computing and SMEs. The classification using open coding, axial coding and selective coding is used to identify the purpose of 50 articles published. In this study context, cluster analysis was applied to assign the articles of each corpus to the most frequently discussed themes or purposes as shown in Figure 1.

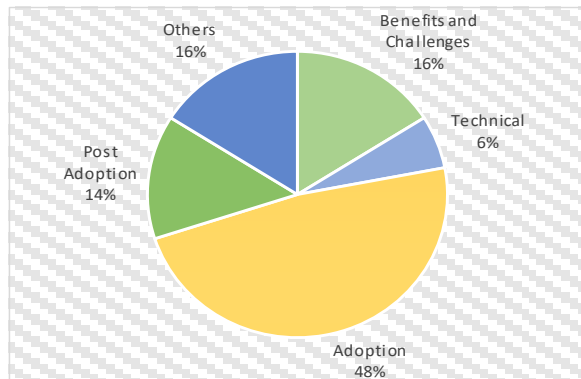


Figure 1 Distribution of studies cloud computing and SMEs.

Figure 1 shows there are five themes identified, namely adoption, benefits and challenges, post-adoption, technical and other issues. The explanation for each theme is as below:

a) Cloud Computing Adoption (48%) - this group contains articles that study intention to adopt, how to adopt and decision to adopt cloud computing by SMEs;

b) Cloud Computing Benefits & Challenges (16%) - this group includes articles which aim to investigate the benefit realization and challenges of cloud computing. The most frequently mentioned benefits include cost saving and pay per use, while for challenges, the common ones include data lock-in, data integrity, security.

c) Other issues (16%) - comprises articles about the organizational, trust, entrepreneurial and issues of cloud computing.

d) Cloud computing Post-Adoption (16%) – contains articles about computing use and impact to SMEs.

e) Technical group of CC (6%) - this group covers articles which scrutinize the architectural design and cloud services such as IaaS, PaaS, SaaS.

In a nutshell, year 2010- 2011 the cloud computing more focus on definition, evolution and taxonomies of CC. However, started year 2012 trend of cloud computing points toward the cloud adoption among SMEs which is 48 %. Then, the trend geared towards in the study of cloud benefits and challenges in 10%. Nowadays, post- adoption has been getting attention among researchers and academicians. Clearly from content analysis there is a huge gap between adoption and post adoption. Therefore, studies on post-adoption of cloud computing still scanty and further research is needed.

B. RQ2: What is the theory, model, framework or concept (TMFC) used in the selected literature?

To response to RQ2, a review was done to identify the theories and models used by past literature related to cloud computing and SMEs. The conceptual approach used for identifying TMFC was adapted from [8]. Table 7 shows the result.

Table 7: Summarize of Theory/Model/Framework/ Concept

Classification	T/M/F/C	Source
Theory	DOI	[17]
	Grounded Theory TAM RBV	[18]
Model	CCBF	[19]
Framework	TOE	[20], [21], [14], [22], [23], [24]
Concept	ATS	[25]

Technology Organization Eevironment (TOE) framework and Diffusion of Innovation (DOI) theory are used most frequently in past researches on cloud computing adoption by [9], [12], [18]–[20]. Review of past literature exposes that cloud computing adoption was investigated by different researchers by considering the TOE framework and DOI theory with the characteristics of innovation. For example, [21] used TOE framework for developing a cloud computing adoption model, that related with the intention of adoption, pricing mechanism, and deployment models among Taiwanese SMEs. On the other hand, Technology Acceptance Model (TAM) was used to predict cloud computing adoption by organizations by [14]. While, a number of recent studies used the RBV to study the impact of cloud computing usage in line with attention of cloud computing post-adoption. Also, the Cloud Computing Business Framework (CCBF) model was suggested to support firms to

attain better in cloud design, deployment, migration and services. The Adaptive Socio-Technical Systems (ASTSs) concept was investigated by [22]. This concept was utilised to provide a framework for cloud computing adoption. This study draws out a set of macro and micro themes connecting to crucial phases of strategy development, the migration process and the impact on customer perceptions.

In conclusion, the results from RQ2 shows that most of the studies have limited their scope by employing a single framework and theoretical perspective, while a few studies combine the DOI and TOE framework. Also, some theories have been used more frequently to understand the cloud computing adoption at individual levels (e.g. TAM) while DOI and RBV theories were applied to understand at organizational level.

C. RQ3 What are the value or benefits of cloud computing for SMEs?

After a scrutiny of the past literature pertinent to cloud computing and SMEs, this section presents a synthesis of cloud adoption value and benefits. There are eight papers that discussed about cloud benefit and SMEs which include [23][24][3][25][7][8][26][15]. The values and benefits identified are grouped together based on related concepts. The results are shown in Table 8.

Table 8: Values and benefits of cloud computing for SMEs

Benefits	Source
Cost Benefit	[30],[28],[14],
Simplicity	[31]
Scalability	[32]
Quick Implementation process	[31]
Disaster Recovery capabilities	[28]

In general, cloud computing's capabilities empower companies to reduce costs, eliminate business frictions and upsurge business agility. Cloud computing also comes with other benefits packages identified as rapid implementation, scalability, quick implementation process and disaster recovery capabilities. Cost benefits of cloud computing provides potential for significant cost reductions (e.g. capital acquisition, IT infrastructure operations, and maintenance cost). Furthermore, cloud computing benefits is rapid implementation. Cloud service provider (CSP) offer wide range of cloud computing solutions, these solutions can satisfy and equip most of the enterprise requirements. Choosing between different solutions and product takes place according to the enterprise business requirements. The implementation process enhanced and accelerate due to this selection process.

Cloud computing offer scalability to cover the fluctuations in the enterprise business size. The scalability choices not only assist to cover the changed business size but at the same time save the costs of unnecessary resources. Cloud computing also

improved accessibility and mobility. Applications over the cloud function in an exposed environment, which rise the accessibility options. The increases accessibility, in turn, increases the usability of the cloud inside the company and outside the enterprise. Moreover, Cloud service provider (CSP) deliver well-defined policies, strategies and plan for restore, recovery and back-up backup that concerned with the availability and disaster recovery.

D. RQ4: What are cloud computing challenges faced by SMEs?

This section presents the discussion on answers related to RQ4 and the finding for cloud computing challenges are shown in Table 9.

Table 9: cloud computing challenges

Cloud computing challenges	Source
Security & Privacy	[32],[33],[14],[23]
Vendor Lock in	[32],
Compliance	[33]
Network	[33]
Market Turbulence	[34]

Table 9 shows most common challenges of cloud computing faced by SMEs. Security and privacy concerns are reported as the key challenge in numerous studies on cloud computing [12], [32], [33]. For example, concerns about losing control of their data and security aspects prevents many SMEs from using cloud computing [29]. In cloud computing, customers are provided access to a shared pool of computing resources over the internet, which further gives rise to concerns regarding the availability of cloud services and its subsequent effect on their industries and business [29]. Another major concern area for SMEs is vendor lock-in, which is clarified as inability or trouble in switching over to other cloud service providers. SMEs also concern regarding poor network consistency and inadequate accessibility of cloud services [16]. In addition, internet connectivity and speed, along with its availability, also affect adoption of cloud computing by SMEs. Then, market turbulence frequently reflects unpredictability, unforeseen and unpredictable in market demands, customers requirements and opponent strategies. Firms in high-turbulence markets face frequent changes in market demands and customer preferences. In this case, SMEs may need to leverage more resources and knowledge for faced market turbulence challenges.

V. CONCLUSION AND FUTURE WORK

This paper presented the results of SLR of recent researches on cloud computing and SMEs which covers published research during the period of 2013-2017. A total of 39 articles were reviewed using the SLR technique to capture the current state of cloud computing adoption among SMEs, theories used, benefits and challenges faced by SMEs. The results show that most past work on cloud computing and SMEs focused on the adoption issue, while only very few explored the impacts or

post adoption stage. This implies that future work may examine the post adoption stage, for example, the challenges involved during implementation, critical success factors and impacts of cloud computing use on SME performance. Additionally, it was found that most past work rely on DOI theory and TOE framework which identify factors for cloud computing adoption. Future work may consider the Resource-based Theory (RBV) to understand how cloud use affect SME performance. This review assist researchers in planning their studies for future work.

VI. ACKNOWLEDGEMENT

This study is part of a research project funded by the Ministry of Education Malaysia FRGS project no FRGS16-008-0507.

VI. REFERENCES

- [1] P. Mell and T. Grance, "The NIST Definition of Cloud Computing Recommendations of the National Institute of Standards and Technology," *Nist Spec. Publ.*, vol. 145, p. 7, 2011.
- [2] A. Khajeh-Hosseini, D. Greenwood, and I. Sommerville, "Cloud migration: A case study of migrating an enterprise IT system to IaaS," *Proc. - 2010 IEEE 3rd Int. Conf. Cloud Comput. CLOUD 2010*, pp. 450–457, 2010.
- [3] M. Carcary, E. Doherty, G. Conway, and S. McLaughlin, "Cloud computing adoption readiness and benefit realization in Irish SMEs-An exploratory study," *Inf. Syst. Manag.*, vol. 31, no. 4, pp. 313–327, 2014.
- [4] B. Kitchenham and S. Charters, "Guidelines for performing Systematic Literature reviews in Software Engineering Version 2.3," *Engineering*, vol. 45, no. 4ve, p. 1051, 2007.
- [5] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in software engineering*, vol. 9783642290. 2012.
- [6] S. Jalali and C. Wohlin, "Systematic Literature Studies: Database Searches vs. Backward Snowballing," *ESEM'12 Proc. ACM-IEEE Int. Symp. Empir. Softw. Eng. Meas.*, pp. 29–38, 2012.
- [7] R. Martins, T. Oliveira, and M. A. Thomas, "An empirical analysis to assess the determinants of SaaS diffusion in firms," *Comput. Human Behav.*, vol. 62, pp. 19–33, 2016.
- [8] E. O. Yeboah-Boateng and K. A. Essandoh, "Factors Influencing the Adoption of Cloud Computing by Small and Medium Enterprises in Developing Economies," *Int. J. Emerg. Sci. Eng.*, vol. 2, no. 4, pp. 13–20, 2014.
- [9] J. N. Makena, "Factors That Affect Cloud Computing Adoption By Small and Medium Enterprises in Kenya," *Int. J. Comput. Appl. Technol. Res.*, vol. 2, no. 5, pp. 517–521, 2013.
- [10] S. A. K. B. Tshiyole and O. Jokonya, "Evaluating the Benefits of Cloud Computing in Small and Medium Enterprises within Gauteng in South Africa," *Int. Conf. Inf. Resour. Manag.*, pp. 1–13, 2016.
- [11] N. Khan and A. Al-Yasiri, "Framework for Cloud Computing Adoption: A Roadmap for Smes to Cloud Migration," *Int. J. Cloud Comput. Serv. Archit.*, vol. 5, no. 5/6, pp. 01–15, 2015.
- [12] S. Al Isma'ili, M. Li, J. Shen, and Q. He, "Cloud computing adoption determinants: an analysis of Australian SMEs Cloud computing adoption determinants: an analysis of Australian SMEs," pp. 1–17, 2016.
- [13] H. Gangwar, H. Date, and R. Ramaswamy, "Understanding determinants of cloud computing adoption using an integrated TAM-TOE model," *J. Enterp. Inf. Manag.*, vol. 28, no. 1, pp. 107–130, 2015.
- [14] T. Oliveira, M. Thomas, and M. Espadanal, "Assessing the determinants of cloud computing adoption: An analysis of the manufacturing and services sectors," *Inf. Manag.*, vol. 51, no. 5, pp. 497–510, Jul. 2014.
- [15] Y. Alshamaila, S. Papagiannidis, and F. Li, "Cloud computing adoption by SMEs in the north east of England: A multi-perspective framework," *J. Enterp. Inf. Manag.*, vol. 26, no. 3, pp. 250–275, 2013.
- [16] J.-W. Lian, "Critical factors for cloud based e-invoice service adoption in Taiwan: An empirical study," *Int. J. Inf. Manage.*, vol. 35, no. 1, pp. 98–109, Feb. 2015.
- [17] B. G. Garrison, S. Kim, and R. L. Wakefield, "Success Factors for Deploying Cloud," *Commun. ACM*, vol. 55, no. 9, pp. 62–68, 2012.
- [18] S. Lee, S. Hoon, and K. Min, "International Journal of Information Management Drivers and inhibitors of SaaS adoption in Korea," *Int. J. Inf. Manage.*, vol. 33, no. 3, pp. 429–440, 2013.
- [19] V. Chang, R. John, and G. Wills, "International Journal of Information Management The development that leads to the Cloud Computing Business Framework," vol. 33, pp. 524–538, 2013.
- [20] F. Safari, N. Safari, and A. Hasanzadeh, "Article information :," 2015.
- [21] S. R. Tehrani and F. Shirazi, "Factors Influencing the Adoption of Cloud Computing by Small and Medium Size Enterprises (SMEs)," pp. 631–642, 2014.
- [22] Y. Y. Alshamaileh, "An Empirical Investigation of factors affecting cloud computing adoption among SMEs in the North," 2013.
- [23] C. Low, Y. Chen, and M. Wu, "Understanding the determinants of cloud computing adoption," *Ind. Manag. Data Syst.*, vol. 111, no. 7, pp. 1006–1023, 2011.
- [24] P. F. Hsu, S. Ray, and Y. Y. Li-Hsieh, "Examining cloud computing adoption intention, pricing mechanism, and deployment model," *Int. J. Inf. Manage.*, vol. 34, no. 4, pp. 474–488, 2014.
- [25] M. Werfs, T. Robert, and I. Sommerville, "Migrating Software Products to the Cloud: An Adaptive STS Perspective," pp. 37–55.
- [26] S. H. Kim, S. Y. Jang, and K. H. Yang, "Analysis of the

- Determinants of Software-as-a-Service Adoption in Small Businesses: Risks, Benefits, and Organizational and Environmental Factors,” *J. Small Bus. Manag.*, vol. 0, no. 0, pp. 1–23, 2016.
- [27] T. Vasiljeva, S. Shaikhulina, and K. Kreslins, “Cloud Computing: Business Perspectives, Benefits and Challenges for Small and Medium Enterprises (Case of Latvia),” *Procedia Eng.*, vol. 178, pp. 443–451, 2017.
 - [28] P. Gupta, A. Seetharaman, and J. R. Raj, “The usage and adoption of cloud computing by small and medium businesses,” *Int. J. Inf. Manage.*, vol. 33, no. 5, pp. 861–874, 2013.
 - [29] G. Garrison, R. L. Wakefield, and S. Kim, “The effects of IT capabilities and delivery model on cloud computing success and firm performance for cloud supported processes and operations,” *Int. J. Inf. Manage.*, vol. 35, no. 4, pp. 377–393, 2015.
 - [30] P. Taylor *et al.*, “Cloud Computing Adoption Readiness and Benefit Realization in Irish SMEs — An Exploratory Study Cloud Computing Adoption Readiness and Benefit Realization in Irish SMEs — An Exploratory Study,” no. March 2015, pp. 37–41, 2014.
 - [31] S. Marston, Z. Li, S. Bandyopadhyay, J. Zhang, and A. Ghalsasi, “Cloud computing - The business perspective,” *Decis. Support Syst.*, vol. 51, no. 1, pp. 176–189, 2011.
 - [32] N. A. Sultan, “Reaching for the ‘cloud’: How SMEs can manage,” *Int. J. Inf. Manage.*, vol. 31, no. 3, pp. 272–278, 2011.
 - [33] S. Gupta and S. C. Misra, “Moderating Effect of Compliance, Network, and Security on the Critical Success Factors in the Implementation of Cloud ERP,” *IEEE Trans. Cloud Comput.*, vol. 4, no. 4, pp. 440–451, 2016.
 - [34] S. Liu, Y. Yang, W. G. Qu, and Y. Liu, “The business value of cloud computing: the partnering agility perspective,” *Ind. Manag. Data Syst.*, vol. 116, no. 6, pp. 1160–1177, 2016.
 - [35] R. D. Raut, B. B. Gardas, M. K. Jha, and P. Priyadarshinee, “Examining the critical success factors of cloud computing adoption in the MSMEs by using ISM model,” *J. High Technol. Manag. Res.*, vol. 28, no. 2, pp. 125–141, 2017.